

Erwin Quanten

Senior DevOps Engineer – AWS DevOps Engineer / Cloud Infrastructure Engineer

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PROFESSIONAL SUMMARY

AWS DevOps Engineer with 16+ years of software engineering and cloud delivery experience across managed services, digital consulting, chemical manufacturing, and enterprise operations, specializing in **AWS Organizations, IAM, VPC, EC2, ALB, S3, RDS, ECS/Fargate, EKS 1.24-1.29, Lambda Node.js 16-20, Terraform 1.3-1.6, Docker 20-25, Kubernetes 1.24-1.29, GitHub Actions v3-v4, Jenkins 2.x, Helm 3.x, CloudWatch, Prometheus 2.x, and Grafana 9-10** to modernize infrastructure, automate releases, reduce deployment risk, strengthen security, improve cost visibility, and support scalable production reliability.

SKILLS

- **Cloud Platforms:** AWS, AWS Organizations, Control Tower, IAM, IAM Identity Center, VPC, EC2, ALB/NLB, Auto Scaling, S3, RDS, Aurora, DynamoDB, ElastiCache, Route 53, CloudFront, Lambda, ECS/Fargate, EKS, SQS, SNS, EventBridge
- **Infrastructure as Code:** Terraform 1.3-1.6, CloudFormation, AWS CDK v2, Ansible 2.x, Packer 1.8-1.10, reusable IaC modules, remote state, drift control
- **CI/CD:** GitHub Actions v3-v4, Jenkins 2.x, AWS CodePipeline, CodeBuild, CodeDeploy, GitLab CI, reusable workflows, approval gates, rollback automation
- **Containers & Orchestration:** Docker 20-25, Kubernetes 1.24-1.29, EKS, ECS, Helm 3.x, Argo CD 2.x, Kustomize, HPA, Cluster Autoscaler
- **Monitoring & Reliability:** CloudWatch, CloudTrail, AWS Config, Prometheus 2.x, Grafana 9-10, OpenSearch, ELK, Datadog, alerting, SLO dashboards, incident response
- **Security & Compliance:** IAM least privilege, KMS, Secrets Manager, Security Hub, GuardDuty, WAF, OIDC, VPC endpoints, audit logging, vulnerability scanning
- **Scripting & Development:** Bash, Python 3.9-3.12, PowerShell 7, Node.js 16-20, YAML, JSON, REST APIs, Linux administration
- **Databases:** PostgreSQL 12-16, MySQL 8, Amazon RDS, Aurora, DynamoDB, Redis, backup, restore, migration, performance tuning

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

Houston, TX

- Architected a multi-account **AWS Organizations** landing zone with **Control Tower, IAM Identity Center, SCPs, VPC 10.x segmentation**, and centralized logging for managed cloud services workloads.
- Moved containerized services from manually maintained EC2 deployments into **ECS/Fargate** and **EKS 1.24-1.29**, improving release stability by **28%** through autoscaling and repeatable deployment templates.
- Built reusable **Terraform 1.3-1.6** modules for VPC, ALB, EKS, RDS, S3, IAM, KMS, and CloudWatch, reducing new environment provisioning effort by **35%** across staging and production accounts.
- Modernized CI/CD pipelines with **GitHub Actions v3-v4, Jenkins 2.x**, OIDC-based AWS authentication, reusable workflow templates, and approval gates that reduced deployment time by **32%**.
- Implemented blue-green and canary deployment patterns using **ALB weighted routing, Route 53**, health checks, and rollback automation, reducing post-release incidents by **22%**.
- Centralized observability with **CloudWatch, Prometheus 2.x, Grafana 9-10**, structured JSON logs, SLO dashboards, and alert routing that shortened production triage by **30%**.
- Hardened secrets and access control with **AWS Secrets Manager, KMS**, short-lived IAM roles, least-privilege policies, and GitHub OIDC to remove long-lived access keys from pipelines.
- Reduced AWS spend by **18%** through rightsizing EC2 and RDS, tuning autoscaling policies, cleaning orphaned EBS volumes, and introducing tagging rules for cost allocation.
- Resolved repeated Terraform state lock and infrastructure drift issues by moving teams to S3 remote state, DynamoDB locking, workspace conventions, and pull-request based plan reviews.
- Fixed EKS production instability caused by pod memory pressure, missing readiness probes, and weak node scaling by tuning HPA, Cluster Autoscaler, resource limits, and deployment probes.
- Implemented backup and disaster recovery standards for **RDS, S3, EBS**, and Kubernetes manifests with tested restore procedures, retention policies, and environment recovery runbooks.

- Mentored developers and DevOps engineers on AWS troubleshooting, secure pipeline design, incident response, and IaC review practices while keeping platform standards simple and reusable.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Led AWS DevOps delivery for digital consulting clients using **Terraform 0.12-1.1**, **Jenkins 2.x**, **Docker 18-20**, **ECS**, and early **EKS 1.14-1.22** adoption across multi-environment platforms.
- Converted manually configured AWS infrastructure into version-controlled Terraform and CloudFormation stacks, reducing configuration drift by **31%** and improving audit readiness.
- Built Jenkins shared libraries, pipeline templates, artifact promotion flows, and automated rollback steps that improved deployment frequency by **26%** without weakening production controls.
- Migrated legacy VM-based Java and Node.js services into **Docker** images hosted on **ECS** and private registries, reducing environment mismatch defects by **24%**.
- Implemented secure network foundations with private subnets, NAT gateways, security groups, ALB rules, VPN access, and VPC endpoints for client-facing enterprise applications.
- Troubleshoot recurring timeout and 502 errors by reviewing ALB target health, container startup probes, JVM memory settings, and connection pool limits across staging workloads.
- Introduced **CloudWatch Logs**, metric filters, alarms, and Grafana dashboards to expose API latency, error rate, CPU, memory, and queue depth before clients reported failures.
- Improved release quality by adding static scans, dependency checks, Docker image scans, and environment smoke tests into CI/CD gates before deployment approval.
- Supported migration from older Jenkins freestyle jobs to Jenkinsfile-based pipelines, preserving release history while standardizing build, test, package, and deploy stages.
- Partnered with product, QA, and security teams to translate business release pressure into practical DevOps automation, clear runbooks, and safer production handover practices.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Supported AWS modernization for chemical manufacturing and enterprise operations using **EC2**, **ELB**, **S3**, **RDS**, **CloudFormation**, **Jenkins 2.x**, **Ansible 2.x**, and controlled Linux automation.
- Moved legacy application deployments from manual server changes to repeatable Jenkins and Ansible workflows, reducing release errors by **23%** across internal business systems.
- Designed CloudFormation templates for EC2, security groups, load balancers, IAM roles, and RDS parameters, improving infrastructure repeatability by **27%** for test and production environments.
- Implemented centralized logging and infrastructure monitoring with **CloudWatch**, **Nagios**, and scripted health checks to improve visibility into slow jobs and unstable services.
- Resolved recurring production defects caused by inconsistent server packages by introducing controlled AMI baselines, configuration scripts, and documented patch procedures.
- Improved database reliability by tuning RDS backup windows, maintenance windows, parameter groups, and restore procedures for applications supporting manufacturing workflows.
- Helped teams migrate from older build scripts to Jenkins 2.x pipeline conventions while keeping deployment steps readable for developers and operations engineers.
- Introduced safer release practices with pre-deployment validation, rollback notes, artifact versioning, and environment checklists that reduced failed releases by **19%**.
- Worked flexibly across development, operations, QA, and business stakeholders to explain AWS changes clearly and keep infrastructure improvements aligned with production needs.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Maintained enterprise application platforms for chemical manufacturing teams using **AWS EC2**, **S3**, **RDS**, **ELB**, **Jenkins 1.x**, **Linux**, **Bash**, and early automation practices.
- Automated deployment packaging and server preparation with Bash and Python scripts, reducing repetitive manual release work by **21%** for internal web applications.
- Supported early migration of selected workloads from on-premise servers to AWS EC2 and S3, improving environment availability by **18%** during business peak periods.
- Configured Jenkins build jobs, artifact archives, test execution, and deployment scripts to help developers detect integration failures before operations handoff.

- Resolved recurring environment issues caused by inconsistent library versions, missing variables, and manual file edits by documenting setup steps and standardizing configuration files.
- Improved operational support by creating health-check scripts, log rotation rules, backup verification tasks, and restart procedures for Linux-hosted business services.
- Worked with database and application teams to tune deployment windows, rollback plans, and SQL script execution so releases were safer for production users.
- Supported Git adoption from older source-control workflows by documenting branch usage, code review expectations, and release tagging conventions for engineering teams.

Tessenderlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Built and supported internal enterprise applications using **Linux, Bash, Apache/Tomcat, Hudson/Jenkins 1.x, Subversion**, early **Git**, and structured release scripts.
- Assisted operations teams with server deployments, log analysis, restart procedures, and configuration checks that reduced repeated support escalations by **16%**.
- Created basic automation scripts for backups, file transfers, build packaging, and environment validation to remove fragile manual steps from routine releases.
- Investigated defects related to missing configuration values, permission errors, and inconsistent server folders, then documented fixes for faster team troubleshooting.
- Collaborated with senior engineers to learn AWS EC2 and S3 fundamentals, CI practices, Linux administration, and practical production support for business applications.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009